

Muhammad Talha Zaheer

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PROFESSIONAL SUMMARY

AI Developer specializing in production agentic systems and real-time voice AI. Builds and ships LLM-powered agents with tool calling, prompt engineering, and speech pipelines (STT/LLM/TTS) over PSTN—including inbound support, outbound SDR, and logistics dispatch agents. Strong full-stack delivery with Python, FastAPI, Next.js, React, and Supabase: agent tools, cost tracking, call analytics, and CI/CD for production, staging, and preview environments. Experienced across Pipecat AI, Retell, OpenAI, Vercel AI SDK, Deepgram, Cartesia, and Daily PSTN; also teaches Agentic AI and applies a security-aware engineering mindset from hands-on cybersecurity training.

CORE COMPETENCIES

AI & Agent Engineering	Agentic AI, LLMs, Tool Calling, Prompt Engineering, System Prompts, Streaming Generation, Voice Agents, Speech Pipelines (STT/LLM/TTS), Cost & Token Tracking, Local LLMs (LM Studio)
AI Frameworks & APIs	Pipecat AI, OpenAI, Vercel AI SDK, Retell AI, Deepgram, Cartesia, Daily PSTN, Twilio
Languages & Backend	Python, TypeScript, JavaScript, SQL, Bash, FastAPI, Node.js, Express, Django
Frontend & Data	Next.js, React, React Native (Expo), Electron, Supabase, WebSockets, PostgreSQL
DevOps & Deployment Tools	GitHub Actions, CI/CD, Docker, Railway, Vercel, Digital Ocean, Git/GitHub, Postman, Cursor, VS Code, Puppeteer, Linear, Monday.com

PROFESSIONAL EXPERIENCE

EPGGB Law	Apr 2026 – Jun 2026
Developer (Contract)	Los Angeles, CA (Remote)

- Built an Electron.js desktop application integrated with the Clio API to fetch, organize, and export legal practice data for client workflows.
- Retrieved and structured records by matters, activities, issues, and bills; mapped API responses into clean, analysis-ready datasets.
- Generated custom-formatted CSV exports for downstream data processing and reporting.

Core Edge Solutions	May 2025 – Apr 2026
AI Engineer / Technical Delivery	Lahore, Pakistan

- Owned end-to-end delivery of production voice-AI products (inbound support, outbound SDR, truck dispatch), coordinating STT/LLM/TTS integrations, backend APIs, and React dashboards from design through release.
- Translated TMS domain requirements into agent prompts, tool specifications, and FastAPI endpoints; aligned engineering work with logistics ops needs (call transfer, email, real-time Trinium lookups).
- Managed release pipelines with GitHub Actions on Railway—production, staging, and per-PR preview environments—reducing deployment friction and enabling safer iteration.
- Tracked delivery quality via per-call cost, recordings, and transcripts for ops review; coordinated Puppeteer/TMS automations that fed live data into agent workflows.
- Used Linear/Monday.com-style task tracking to prioritize features, surface blockers, and keep multi-component agent platforms on schedule.

Professional Freelancing Training Program	Nov 2024 – Apr 2025
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- Supported security-focused project work through structured testing, documentation of findings, and remediation recommendations (web app testing, network analysis, WAF/firewall configuration).

Freelance

Sep 2023 – Nov 2024

Freelance Developer

Lahore, Pakistan

- Delivered freelance software projects for restaurant clients—offline POS systems, waiter mobile apps, and invoice/billing desktop applications built with Node.js and Electron.
- Owned requirements gathering, database design, implementation, and client delivery across hybrid engagements from scoping through deployment.
- Contributed to open projects involving database architecture design and cybersecurity workflow development.

PROJECTS

Vello (Voice Agent)

AI Agent

Tech Stack: Pipecat AI, Deepgram STT, OpenAI LLM, Cartesia TTS, Daily PSTN, Supabase, React

- Built a Pipecat AI voice bot for inbound and outbound calls via Daily PSTN, with voicemail detection, call transfer, and professional sales-agent conversations.
- Integrated Deepgram STT, OpenAI LLM, and Cartesia TTS; supports separate dynamic system prompts for inbound and outbound agents loaded from Supabase at call time.
- Tracks token consumption for per-call pricing; stores recordings and transcriptions in Supabase with a React dashboard for org setup, member invites, recordings, and cost analytics.

Pacifica Bot (Customer Support Agent)

Voice Agent

Tech Stack: Pipecat AI, Deepgram STT, OpenAI LLM, Cartesia TTS, Daily PSTN, FastAPI, Supabase, React, GitHub Actions, Railway

- Built an inbound Pipecat AI voice agent via Daily PSTN for logistics customer support, with a comprehensive system prompt covering general guidelines and TMS domain knowledge to resolve queries efficiently, including generic inquiries.
- Implemented agent tools as FastAPI endpoints that query the Trinium database with optimized joins for fast real-time lookups; supports ending calls, transferring calls, and sending emails during live conversations.
- Tracks per-call cost across LLM, TTS, STT, and telephony usage; stores call recordings and transcripts in Supabase for later analysis and review.
- Designed GitHub Actions workflows to deploy on Railway with separate pipelines for production, staging, and per-PR preview environments.

Zuum (Truck Dispatch Agent)

Voice Agent

Tech Stack: Retell AI, Twilio, FastAPI, React, Supabase, WebSockets

- Built a truck dispatch voice agent using Retell AI and Twilio to outbound-call drivers for load status updates, delivery confirmation, hurdle reporting, and POD collection.
- Implemented Retell single-prompt agents with custom functions to fetch dispatch data from a FastAPI backend; used post-call analysis and webhook events to track live and completed calls.
- Stored call data, analysis, and transcriptions in Supabase; React frontend displays dispatch activity with real-time updates via WebSockets.

Automated SDR (Outbound Calling Agent)

Voice Agent

Tech Stack: Pipecat AI, Deepgram STT, OpenAI LLM, Cartesia TTS, Daily PSTN, Supabase, React, GitHub Actions, Railway

- Built an outbound Pipecat AI voice agent for sales development, placing automated calls for prospecting, lead qualification, and follow-up at scale via Daily PSTN.
- Integrated Deepgram STT, OpenAI LLM, and Cartesia TTS with dynamic SDR prompts loaded from

Supabase; includes voicemail detection and professional sales conversation flows.

- Tracks token usage for per-call cost analysis; stores recordings and transcriptions in Supabase with a React dashboard for campaign monitoring and call review.
- Designed GitHub Actions workflows to deploy on Railway with separate pipelines for production, staging, and per-PR preview environments.

Agentic Portfolio

AI Web App

Tech Stack: Next.js, Vercel AI SDK, OpenAI, Vercel

- Built a Next.js portfolio app powered by the Vercel AI SDK and OpenAI, using a user-supplied JSON CV as dynamic context for the chatbot.
- UI adapts automatically from the uploaded JSON profile; the agent answers role-fit and project-related queries from that resume data.
- Designed for easy deployment on Vercel with minimal setup—users only need to provide their CV JSON to launch a personalized agentic portfolio.

DrayDog – Puppeteer Automations

Automation Platform

Tech Stack: Node.js, Express, Puppeteer, OpenAI

- Built a Node.js automation platform that fetches container data from DrayDog and runs concurrent Puppeteer workflows for logistics operations.
- Automated marking containers as picked up, updating leg movement status, adding driver notes, and flagging empty containers as available for pickup.
- Collected and stored DrayDog and Trinium data for cross-system comparison; used OpenAI to analyze empty-container emails.

AstroPOS

Restaurant Management System

Tech Stack: Electron, Vite, React, Supabase

- Built a complete restaurant management system with PIN login, multi-kitchen setup, menu items, deals, custom shift tracking, and per-item sales reporting.
- Supports multiple payment accounts, on-account customer billing, auto-saved customer details, delivery charges, and discounts via a full admin panel.
- Works offline and online with interval-based sync; routes orders to multiple printers with separate customer and kitchen receipts; includes an on-screen keyboard and auto-update support.

Chotu Cafe Mobile

Mobile App

Tech Stack: Expo, React Native, Supabase

- Built a waiter-facing mobile app in Expo with Supabase for managing table orders across restaurant service workflows.
- Uses Supabase Realtime to push live order updates as changes are made or new orders are placed.
- Sends remote print instructions via a print server to route receipts to kitchens and customers when needed.

Astro CTF

CTF Platform

Tech Stack: Django, PostgreSQL, Docker, Jinja2

- Built a complete CTF platform in Django with modular apps, PostgreSQL backend, and event lifecycle management (start/end scheduling).
- Supports tasks with encrypted flags, files, links, hints, and Docker-based challenges added via a guided admin UI.
- Tracks team and per-task performance metrics; uses Jinja2 templating to render structured frontend views from large datasets.

Darzee (Tailor Marketplace) — Final Year Project

Mobile App

Tech Stack: Kotlin, Jetpack Compose, Android Studio, Django REST Framework, PostgreSQL, Bootstrap, Git

- Built a mobile app connecting tailors and customers for custom clothing orders, order progress tracking, nearest-tailor discovery, and service quotes.

- Implemented tailor, customer, and order management with Android (Kotlin, Jetpack Compose) and a Django REST Framework + PostgreSQL backend.

File Sync System

C++ Application

Tech Stack: C++, WinSock, AVL Tree

- Built separate C++ client and server applications on Windows using WinSock for networked file management.
- Server spawns a dedicated thread per client connection and uses an in-memory AVL tree to index millions of files for fast lookup.
- Client supports account login and Linux-style commands to list, create, delete, and modify files on the remote server.

Portfolio: talhazaheer.dev/#projects

EDUCATION

University of Management and Technology (UMT)

Bachelor of Science, Computer Science

Nov 2019 – Aug 2025

Lahore, Pakistan

Professional Freelancing Training Program (NIAIS)

Diploma, Cyber Security

Jun 2024 – May 2025

Lahore, Pakistan

NAVTTC

Certification, Digital Forensic & Cyber Security

Apr 2025 – Jul 2025

Pakistan

ADDITIONAL EXPERIENCE

Hack The Box

Meetup Host & Ambassador

Apr 2025 – Present

Lahore, Pakistan

Hosted on-campus and online cybersecurity meetups at NIIT, UCP, FAST Lahore, UMT Lahore, Superior University, ShellShift, and virtual sessions, teaching security fundamentals through workshops and hands-on labs.

Skill Bridge Pakistan

Agentic AI Course Instructor

Dec 2025 – May 2026

Lahore, Pakistan

Recorded 24 lectures and guided cohorts through LLM fundamentals, prompt engineering, Vercel AI SDK agents, and deploying an agentic portfolio on Vercel—planning curriculum milestones and student deliverables.

Hunarmand Punjab

Course Instructor

Sep 2025 – Oct 2025

Punjab, Pakistan

Delivered structured technical training for a government skills initiative; designed and recorded two courses (Cybersecurity and Penetration Testing, 48 lectures) with quizzes, assignments, and exams.